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THIRD SEMESTER B. TECH EXAMINATION, DECEMBER-2024

FLUID MECHANICS AND MACHINERY

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 X 2=10**		
1.	**a.**	Describe the phenomenon of fluid a continuum with an example.	L2	CO1	**02**
	b	Explain the vacuum pressure and atmospheric pressure.	L2	CO2	**02**
	c.	Explain Chezy's formula for calculating the loss of head due to friction in pipes and provide the formula.	L2	CO3	**02**
	d	What is the basic working principle of hydraulic machines?	L2	CO4	**02**
	e.	List any two types of heads in a pump.	L2	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain the roles of adhesive and cohesive forces in fluid mechanics. Give examples	L2	CO1	**05**
		OR			
	b.	State and prove Pascal's law.	L2	CO1	**05**
3.	**a.**	If the velocity distribution over a plate is given by $u= (2/3)\ y-y^2$ in which u is the velocity in meter per second at a distance y meter above the plate, determine the shear stress at y=0 and y=0.15m. Take dynamic viscosity of fluid as 8.63 poise.	L2	CO1	**05**
		OR			
	b.	State and Prove continuity equation in two dimensions	L2	CO1	**05**
4.	**a.**	Distinguish between the following types of fluid flow with examples, **i.** Steady and Unsteady flow **ii.** Rotational and Irrotational flow	L3	CO2	**05**
		OR			
	b	A hydraulic press has a ram of 20 cm diameter and a plunger of 3 cm diameter. It is used for lifting a weight of 30kN. Find the force required at the plunger	L3	CO2	**05**
5.	**a.**	Derive an expression for capillary fall in a fluid .	L3	CO3	**05**
		OR			
	b	Explain the loss in pipe due to sudden contraction	L2	CO3	**05**

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6.	a.	What are draft tubes? Explain any two of them with sketches	L2	CO4	05
		OR			
	b	Discuss the importance of governing of turbines in maintaining the stability of power generation in response to load variations.	L2	CO4	5
7.	a.	Explain the working of reciprocating pump with neat sketch	L2	CO5	5
		OR			
	b	Explain the working of centrifugal pump with a neat sketch	L2	CO5	5
		PART-C	**6 X 10=60**		
8.	a.	Explain Newton's law of viscosity in detail. Derive the mathematical expression for Newton's law of viscosity and discuss its significance in fluid mechanics.	L3	CO1	10
		OR			
	b	What are the guage pressure and absolute pressure at a point 3 m below the free surface of a liquid having a density of $1.53X10^3$ kg/m^3 if the atmospheric pressure is equivalent to 750mm of mercury ? The specific gravity of mercury is 13.6 and density of watSer $=1000kg/m^3$	L3	CO1	10
9.	a.	State and prove Hydrostatic law	L3	CO1	10
		OR			
	b	A U-tube manometer is used to measure the pressure of oil of specific gravity 0.85 flowing in a pipe line. Its left end is connected to the pipe and the right-limb is open to the atmosphere. The centre of the pipe is 100 mm below the level of mercury (specific gravity = 13.6) in the right limb. If the difference of mercury level in the two limbs is 160 mm, determine the absolute pressure of the oil in the pipe.	L3	CO1	10
10	a.	Derive an expression for coefficient of discharge for venturimeter.	L3	CO2	10
		OR			
	b	Water is flowing through a pipe having diameters 600 mm and 400 mm at the bottom and upper end respectively. The intensity of pressure at the bottom end is 350 kN/m^2 and the pressure at the upper end is 100 kN/m^2. Determine the difference in datum head if the rate of flow through the pipe is 60 litre/sec	L3	CO2	10

11	**a.**	Describe Rayleigh's method to formulate a dimensionally homogeneous equation between the various physical quantities effecting a certain phenomenon.	L2	CO3	**10**
		OR			
	b	Find the diameter of a pipe of length 200m when the rate of flow through the pipe is 200 liters/second and the head lost due to friction is 4. Take the value of C =50 in Chezys formula	L3	CO3	**10**
12	**a.**	i. State causes of cavitation in hydraulic turbine and methods of avoiding it. ii. State advantages and disadvantages of Francis turbine over Pelton wheel.	L2	CO4	**10**
		OR			
	b	Explain criteria for the selection of hydraulic turbines and discuss hydraulic efficiency, mechanical efficiency and overall efficiency with expression.	L2	CO4	**10**
13		Derive an expression for the pressure rise in the impeller of a centrifugal pump by neglecting the frictional and other losses in the impeller.	L3	CO5	**10**
		OR			
	b	Explain the following terms: Net positive suction head, priming and cavitation.	L2	CO5	**10**